

# TrES-1b

R-2646

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-1\_260725 · created 2026-08-29T22:54:37+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2461246.7976 +/- 0.0034 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.183 +/- 0.054                 |
| Transit depth (Rp/Rs)^2                           | 3.35 +/- 1.98 [%]               |
| Orbital Inclination (inc)                         | 89.82 +/- 1.37                  |
| Airmass coefficient 1 (a1)                        | 1.24 +/- 0.099                  |
| Airmass coefficient 2 (a2)                        | -0.2 +/- 0.14                   |
| Scatter in the residuals of the lightcurve fit is | 7.88 %                          |
| Best Comparison Star                              | #3 - [290, 207]                 |
| Optimal Aperture                                  | 2.79                            |
| Optimal Annulus                                   | 19.05                           |
| Transit Duration (day)                            | 0.0881 +/- 0.0046               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.183 $\pm$ 0.054 vs 0.1358 (deviation 0.5 $\sigma$ ) |
| Duration fit vs published            | 0.0881 d vs 0.1042 d (deviation 1.99 $\sigma$ )       |
| Mid-time O-C                         | 7.0 min                                               |
| Depth SNR                            | 1.9                                                   |
| Residual scatter                     | 7.88 %                                                |

### Light curve

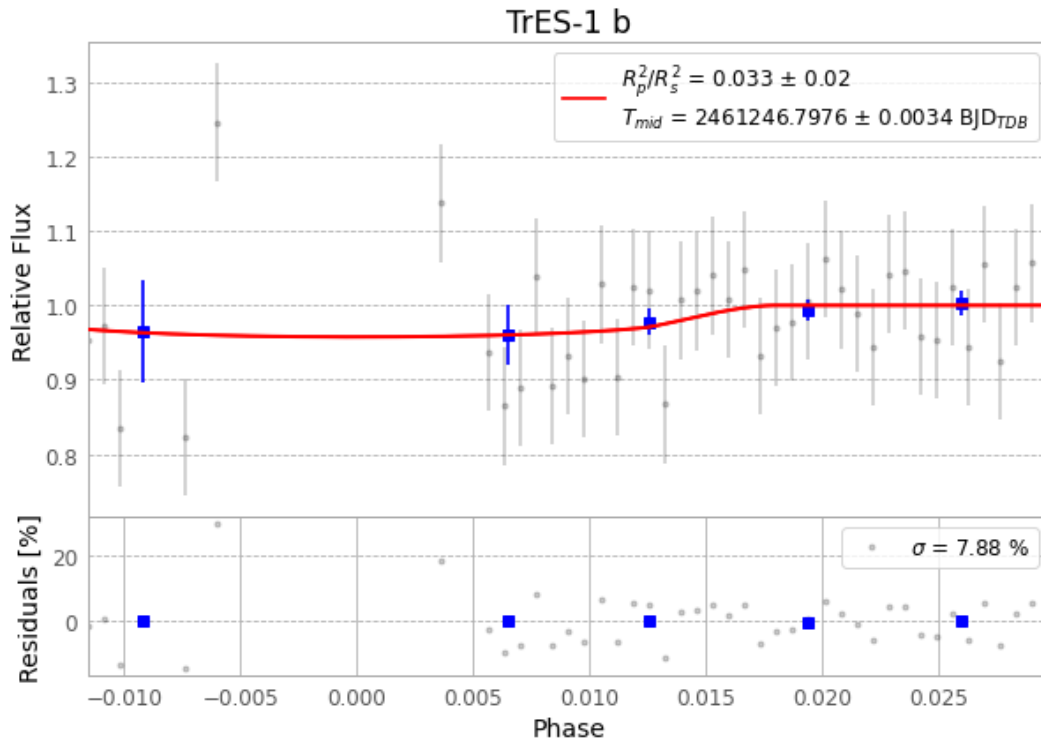


Figure 1 — FinalLightCurve\_TrES-1 b\_2026-07-24.png (SHA-256 32eb44273c6345cf...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 200], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 857bd89e0f776050...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

61 FITS frames (40 MB), TRES-1260725061215.FITS → TRES-1260725091216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-260725.log (55c84d7e1382...), FinalLightCurve\_TrES-1 b\_2026-07-24.png (32eb44273c63...), FinalLightCurve\_TrES-1 b\_2026-07-24.csv (dca12f0680cc...)

## Compute resources

Wall time 179.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 22:54Z.